Year: 2026-2027 **Sem V-NEP**

Course: Agile Software Development and DevOps Lab

Experiment 02 - LO4

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Class: T1

Student Roll No: 2401044 **Student Name:Bhavisha Kewalramani**

LO3: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Aim: To build, package, and containerize a simple web application using Docker and run it inside a Docker container.

Theory

1. Containerization vs. Virtualization

Containerization is an OS-level virtualization method that allows applications to run in isolated user spaces called **containers**. Unlike traditional Virtual Machines (VMs) that require a full hypervisor and guest operating system, containers share the host operating system's kernel. This makes them significantly lightweight, fast to start, and resource-efficient.

2. Docker Architecture

Docker is an open-source platform that automates application deployment inside containers. Key architecture components include:

- **Docker Client:** The command-line interface (docker CLI) used to issue commands.
- **Docker Daemon (dockerd):** The background service that manages Docker objects (images, containers, networks, and volumes).
- **Docker Image:** A read-only template with instructions and dependencies for creating a container.
- **Docker Container:** A lightweight, runnable instance of a Docker image.
- **Docker Registry:** A repository (like Docker Hub) used to store and pull container images.

3. Key Commands Summary

- **Image Operations:** docker pull (download image), docker images (list local images).
- **Container Lifecycle:** docker run (create and start), docker stop (halt container), docker rm (delete container).
- **Monitoring & Inspection:** docker ps (list active containers), docker logs (view output), docker container stats (view CPU/RAM usage).
- **Interactive Access:** docker exec -it <container> bash (open shell inside container).

Steps to install Docker:-

Before installing, ensure your system meets these requirements:

- **OS: Windows 10/11 64-bit (Home, Pro, Enterprise, or Education).**
- **Virtualization: Enabled in system BIOS/UEFI.**
- **WSL 2: Recommended for optimal performance on Windows.**

Installation Guide

Step 1: Download Docker Desktop

1. Visit the official download page: Docker Desktop Download Page.
2. Click Download for Windows to get the installer file (Docker Desktop Installer.exe).

Step 2: Run the Installer

1. Double-click Docker Desktop Installer.exe to start the setup.
2. When prompted, ensure the following options are checked:
    - Use WSL 2 instead of Hyper-V *(recommended for performance and resource sharing)*.
    - Add shortcut to desktop *(optional)*.
3. Click Ok and wait for the installer to extract files and complete.

Step 3: Restart Your Computer

1. Once installation finishes, click Close and restart to apply system changes.

Step 4: Launch & Finalize Setup

1. After rebooting, launch Docker Desktop from your Start menu or desktop shortcut.
2. Accept the Docker Subscription Service Agreement.
3. If prompted to update/install the WSL 2 Linux kernel package:
    - Open PowerShell as Administrator and run: wsl --install
    - Restart your PC if prompted by the WSL installer.

Step 5: Verify the Installation

Open Command Prompt (CMD) or PowerShell and verify everything works:

1. Check Docker Version:

   docker --version
2. Run a Test Container:

   docker run hello-world

If set up properly, Docker will pull the test image and print a *"Hello from Docker!"* confirmation message right in your terminal.

List of Commands executed:

1. Host System Commands

- **Verification & Image Operations**
  - docker --version
  - docker images
  - docker search ubuntu --filter stars=1000
  - docker pull ubuntu
- **Container Launching & Management**
  - docker run -it --name my_ubuntu_app ubuntu
  - docker ps -a
  - docker start my_ubuntu_app
  - docker ps
  - docker ps -a
  - docker stop my_ubuntu_app
  - docker rm my_ubuntu_app
- **Detached Containers & Inspection**
  - docker run -d --name my_bg_container ubuntu sleep 300
  - docker inspect my_bg_container
  - docker exec -it my_bg_container bash *(Executed twice)*
- **Port Mapping, Logging & Monitoring**
  - docker run -d --name my_web_server -p 8080:80 nginx
  - docker logs my_web_server
  - docker container stats
  - docker run -d -p 8080:80 --name my_web_server nginx

2. Inside-Container Commands

- **Inside** my_ubuntu_app**:**
  - cat /etc/os-release
  - echo "Containerization Experiment Success!" > test.txt
  - cat test.txt
  - exit
- **Inside** my_bg_container **(via** docker exec**):**
  - hostname
  - exit

Commands executed:-

docker --version *(Verification)*

docker images (Local repository check)

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\15L>docker images
REPOSITORY                TAG       IMAGE ID       CREATED        SIZE
nginx                     latest    4e5db4761e0f   5 days ago     161MB
ubuntu                    latest    de7345b16e94   7 days ago     100MB
nginx                     <none>    9d0e6f6199dc   8 months ago   152MB
welcome-to-docker         latest    def189b16f2a   2 years ago    226MB
<none>                    <none>    0e7372b5f723   2 years ago    226MB
ubuntu                    <none>    ca2b0f26964c   2 years ago    77.9MB
mysql                     8.3.0     019814493c7a   2 years ago    632MB
mysql                     latest    019814493c7a   2 years ago    632MB
redis                     latest    170a1e90f843   2 years ago    138MB
docker/welcome-to-docker  latest    c1f619b6477e   2 years ago    18.6MB
nginx/docker-extension    0.0.3     41d3d0d7d940   3 years ago    7.53MB
```

docker search ubuntu --filter stars=1000 *(Docker Hub filtering)*

```
C:\Users\15L>docker search ubuntu --filter stars=1000
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
```

docker pull ubuntu *(Image download)*

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
```

```
C:\Users\15L>docker images
REPOSITORY                TAG       IMAGE ID       CREATED        SIZE
nginx                     latest    4e5db4761e0f   5 days ago     161MB
ubuntu                    latest    de7345b16e94   7 days ago     100MB
nginx                     <none>    9d0e6f6199dc   8 months ago   152MB
welcome-to-docker         latest    def189b16f2a   2 years ago    226MB
<none>                    <none>    0e7372b5f723   2 years ago    226MB
ubuntu                    <none>    ca2b0f26964c   2 years ago    77.9MB
mysql                     8.3.0     019814493c7a   2 years ago    632MB
mysql                     latest    019814493c7a   2 years ago    632MB
redis                     latest    170a1e90f843   2 years ago    138MB
docker/welcome-to-docker  latest    c1f619b6477e   2 years ago    18.6MB
nginx/docker-extension    0.0.3     41d3d0d7d940   3 years ago    7.53MB
```

docker run -it --name my_ubuntu_app ubuntu ***(Container launch)***

Inside the container: cat /etc/os-release, echo ... > test.txt, cat test.txt, exit

```
C:\Users\15L>docker run -it --name my_ubuntu_app ubuntu
root@63c2571a7e95:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

```
root@63c2571a7e95:/# echo "Containerization Experiment Success!" > test.txt
cat test.txt
Containerization Experiment Success!
root@63c2571a7e95:/#
```

docker ps -a ***(List all containers including stopped)***

```
root@63c2571a7e95:/# exit
exit

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED              STATUS
                    PORTS                  NAMES
63c2571a7e95   ubuntu                               "/bin/bash"              About a minute ago   Exited
(0) 5 seconds ago                         my_ubuntu_app
b4e15dfed6bf   nginx                                "/docker-entrypoint.…"   18 hours ago         Exited
(0) 18 hours ago                          busy_lovelace
4b486d8501d7   ubuntu                               "/bin/bash"              19 hours ago         Exited
(127) 19 hours ago                        myContainer
b15df8ca655b   ca2b0f26964c                         "/bin/bash"              19 hours ago         Exited
(0) 19 hours ago                          sleepy_yonath
79d8deb2f28e   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago         Created
                                          ecstatic_satoshi
065e20c61331   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago         Exited
(255) 19 hours ago   0.0.0.0:8000->80/tcp  keen_morse
4e2f1b0ebad1   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago         Exited
(0) 8 months ago                          angry_torvalds
578a8f30694f   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago         Created
                                          relaxed_jepsen
4b2344abd78a   mysql:latest                         "docker-entrypoint.s…"   2 years ago          Exited
(1) 2 years ago                           practical_spence
66c4141a85c5   mysql:8.3.0                          "docker-entrypoint.s…"   2 years ago          Exited
(0) 2 years ago                           mydb
0a805a3dc56c   ca2b0f26964c                         "/bin/bash"              2 years ago          Exited
(0) 2 years ago                           vibrant_lamarr
7fec2031b676   0e7372b5f723                         "docker-entrypoint.s…"   2 years ago          Exited
(0) 2 years ago                           trusting_newton
df0f11f819ff   docker/welcome-to-docker:latest      "/docker-entrypoint.…"   2 years ago          Exited
(137) 2 years ago                         welcome-to-docker
```

docker start my_ubuntu_app ***(Start stopped container)***

docker ps ***(List running containers)***

```
C:\Users\15L>docker start my_ubuntu_app
my_ubuntu_app

C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS          PORTS     NAMES
63c2571a7e95   ubuntu    "/bin/bash"   2 minutes ago   Up 11 seconds             my_ubuntu_app

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS
               PORTS                    NAMES
63c2571a7e95   ubuntu                               "/bin/bash"              3 minutes ago   Up 41 second
s                                       my_ubuntu_app
b4e15dfed6bf   nginx                                "/docker-entrypoint.…"   18 hours ago    Exited (0) 1
8 hours ago                             busy_lovelace
4b486d8501d7   ubuntu                               "/bin/bash"              19 hours ago    Exited (127)
 19 hours ago                           myContainer
b15df8ca655b   ca2b0f26964c                         "/bin/bash"              19 hours ago    Exited (0) 1
9 hours ago                             sleepy_yonath
79d8deb2f28e   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Created
                                        ecstatic_satoshi
065e20c61331   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Exited (255)
 19 hours ago   0.0.0.0:8000->80/tcp   keen_morse
4e2f1b0ebad1   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Exited (0) 8
 months ago                             angry_torvalds
578a8f30694f   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Created
                                        relaxed_jepsen
4b2344abd78a   mysql:latest                         "docker-entrypoint.s…"   2 years ago     Exited (1) 2
 years ago                              practical_spence
66c4141a85c5   mysql:8.3.0                          "docker-entrypoint.s…"   2 years ago     Exited (0) 2
 years ago                              mydb
0a805a3dc56c   ca2b0f26964c                         "/bin/bash"              2 years ago     Exited (0) 2
 years ago                              vibrant_lamarr
7fec2031b676   0e7372b5f723                         "docker-entrypoint.s…"   2 years ago     Exited (0) 2
 years ago                              trusting_newton
df0f11f819ff   docker/welcome-to-docker:latest      "/docker-entrypoint.…"   2 years ago     Exited (137)
 2 years ago                            welcome-to-docker
```

docker stop my_ubuntu_app ***(Stop running container)***

docker rm my_ubuntu_app ***(Delete container)***

```
C:\Users\15L>docker stop my_ubuntu_app
my_ubuntu_app

C:\Users\15L>docker rm my_ubuntu_app
my_ubuntu_app

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS
               PORTS                    NAMES
b4e15dfed6bf   nginx                                "/docker-entrypoint.…"   18 hours ago    Exited (0) 18
 hours ago                              busy_lovelace
4b486d8501d7   ubuntu                               "/bin/bash"              19 hours ago    Exited (127)
19 hours ago                            myContainer
b15df8ca655b   ca2b0f26964c                         "/bin/bash"              19 hours ago    Exited (0) 19
 hours ago                              sleepy_yonath
79d8deb2f28e   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Created
                                        ecstatic_satoshi
065e20c61331   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Exited (255)
19 hours ago   0.0.0.0:8000->80/tcp   keen_morse
4e2f1b0ebad1   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Exited (0) 8
months ago                              angry_torvalds
578a8f30694f   9d0e6f6199dc                         "/docker-entrypoint.…"   8 months ago    Created
                                        relaxed_jepsen
4b2344abd78a   mysql:latest                         "docker-entrypoint.s…"   2 years ago     Exited (1) 2
years ago                               practical_spence
66c4141a85c5   mysql:8.3.0                          "docker-entrypoint.s…"   2 years ago     Exited (0) 2
years ago                               mydb
0a805a3dc56c   ca2b0f26964c                         "/bin/bash"              2 years ago     Exited (0) 2
years ago                               vibrant_lamarr
7fec2031b676   0e7372b5f723                         "docker-entrypoint.s…"   2 years ago     Exited (0) 2
years ago                               trusting_newton
df0f11f819ff   docker/welcome-to-docker:latest      "/docker-entrypoint.…"   2 years ago     Exited (137)
2 years ago                             welcome-to-docker
```

Run a container in detached mode (background)

docker run -d --name my_bg_container ubuntu sleep 300

```
C:\Users\15L>docker run -d --name my_bg_container ubuntu sleep 300
d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f
```

Inspect a running or stopped container

docker inspect my_bg_container



```
C:\Users\15L>docker inspect my_bg_container
[
    {
        "Id": "d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f",
        "Created": "2026-07-21T05:17:46.840654725Z",
        "Path": "sleep",
        "Args": [
            "300"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 1016,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-21T05:17:47.080539925Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:de7345b16e942e22044c6ba053020ec85ae879984860a9918517d54eb6cef851",
        "ResolvConfPath": "/var/lib/docker/containers/d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f/hostname",
        "HostsPath": "/var/lib/docker/containers/d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f/hosts",
        "LogPath": "/var/lib/docker/containers/d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f/d7af37520f1100c00fbda87e5361268108f2d84f12afbc6842a5d82708757f4f-json.log",
        "Name": "/my_bg_container",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "default",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
```

Open a shell inside a running container

docker exec -it my_bg_container bash

```
C:\Users\15L>docker exec -it my_bg_container bash
root@d7af37520f11:/#
```

```
C:\Users\15L>docker exec -it my_bg_container bash
root@d7af37520f11:/# hostname
d7af37520f11
root@d7af37520f11:/# exit
exit
```

Viewing Container Resource Usage Statistics

docker run -d -p 8080:80 --name my_web_server nginx

```
C:\Users\15L>docker run -d --name my_web_server -p 8080:80 nginx
9eb870b2ae7abe2f1fae9e9f35c49bb6f4a6c10715e5552d17dc8e7320a77e2c

C:\Users\15L>docker logs my_web_server
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/21 05:20:58 [notice] 1#1: using the "epoll" event method
2026/07/21 05:20:58 [notice] 1#1: nginx/1.31.3
2026/07/21 05:20:58 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/21 05:20:58 [notice] 1#1: OS: Linux 5.15.133.1-microsoft-standard-WSL2
2026/07/21 05:20:58 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/21 05:20:58 [notice] 1#1: start worker processes
2026/07/21 05:20:58 [notice] 1#1: start worker process 29
2026/07/21 05:20:58 [notice] 1#1: start worker process 30
2026/07/21 05:20:58 [notice] 1#1: start worker process 31
2026/07/21 05:20:58 [notice] 1#1: start worker process 32
2026/07/21 05:20:58 [notice] 1#1: start worker process 33
2026/07/21 05:20:58 [notice] 1#1: start worker process 34
2026/07/21 05:20:58 [notice] 1#1: start worker process 35
2026/07/21 05:20:58 [notice] 1#1: start worker process 36
2026/07/21 05:20:58 [notice] 1#1: start worker process 37
2026/07/21 05:20:58 [notice] 1#1: start worker process 38
2026/07/21 05:20:58 [notice] 1#1: start worker process 39
2026/07/21 05:20:58 [notice] 1#1: start worker process 40
2026/07/21 05:20:58 [notice] 1#1: start worker process 41
2026/07/21 05:20:58 [notice] 1#1: start worker process 42
2026/07/21 05:20:58 [notice] 1#1: start worker process 43
2026/07/21 05:20:58 [notice] 1#1: start worker process 44
2026/07/21 05:20:58 [notice] 1#1: start worker process 45
2026/07/21 05:20:58 [notice] 1#1: start worker process 46
2026/07/21 05:20:58 [notice] 1#1: start worker process 47
2026/07/21 05:20:58 [notice] 1#1: start worker process 48
```

5. View resource usage stats

docker container stats

```
C:\Users\15L>docker container stats
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
^C
C:\Users\15L>
```

6. Publish ports to host (Port Mapping)

docker run -d -p 8080:80 --name my_web_server nginx

```
C:\Users\15L>docker run -d -p 8080:80 --name my_web_server nginx
7e3e45492e51a34ce40533707525ad037723de0a5e3cd05ee620c40eae5d15a1
```

Conclusion

In this experiment, I got hands-on experience containerizing applications and running them using Docker Desktop on Windows.

Key Skills & Learnings:

- **Handling Images and Containers:** I learned how to download base images from Docker Hub with docker pull, check my local image repository using docker images, and launch new containers with docker run.
- **Managing Container Lifecycles:** I practiced starting, stopping, and removing containers (docker start, docker stop, docker rm) to keep my workspace clean and free up host resources.
- **Running Modes & Terminal Access:** I worked with detached mode (-d) to keep services running in the background, and used interactive mode (-it) alongside docker exec to jump into container terminal sessions and execute commands directly inside them.
- **Troubleshooting & Monitoring:** I used diagnostic tools like docker ps -a to track container statuses, docker inspect to check container setups, docker logs to view real-time log outputs, and docker container stats to monitor live CPU and RAM usage.

Overall, this lab helped me understand how Docker isolates dependencies and makes running applications consistent across different environments without the heavy overhead of traditional virtual machines.